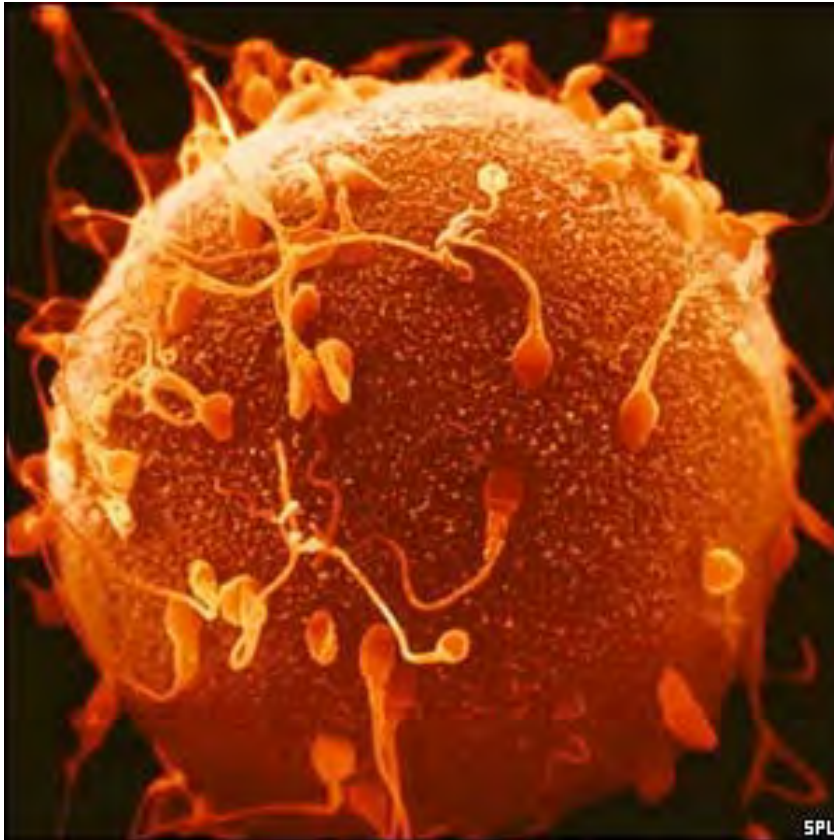


Bio 40C: Reproduction

Jeff Schinske



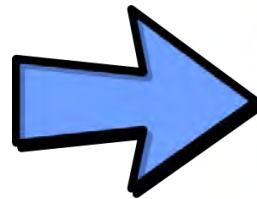


Bio 40C Proposed Norms

- Seek to enjoy your colleagues and your experiences in class!
- Actively select language that is inclusive, respectful, and welcoming
- Assume good intentions...no heroes and no villains (in this class)
- Recognize that we all have strengths as well as areas for growth
- Be curious, be skeptical
- Ask for help (and expect to receive it!)

Share with a partner...

1. Name
2. Something important to know about you as a person
3. What plans do you have for the holiday weekend ahead?



Person with 1st name closest to start of alphabet goes first



Debriefing Last Week

○ Anonymous End of Class Index Cards

- "Something that stood out to me were the memory T cells. It's cool how they're able to remember antigens for so long when I can't even remember what I had for breakfast."
- "I have a friend who has lupus, so I feel I have a better understanding of how autoimmunity is happening inside his body."
- "I enjoyed the case studies and being able to check on diagnoses and diagrams with teammates."

○ Reflecting On Canvas

- "I really enjoyed this week's lesson. I believe every person should understand what's going on with human reproduction and I admittedly did not know a lot of information that was given."
- "This has been a long week for me and a lot of the material was extremely dense. I am feeling very tired and burnt out especially with the holidays coming up."



A Brief Preview of the Week

- Today
 - Review of Application Question 12.
 - Considering male reproductive physiology through the lens of contraception.
- Lab This Week
 - Reproductive System
- Online Lecture Work This Week
 - None due to holiday weekend!
 - Just a prelab quiz for next week's lab meeting.



Our Specific Outcomes

- Today you will be able to...
 - Relate human reproductive physiology and hormonal feedback loops to chemical contraception.
 - Demonstrate the significance of anatomical structures in the testes to issues of fertility/infertility.



A Reminder of Our Disclaimer...

- “Male” will refer to the most common A&P of humans that produce sperm.
- “Female” will refer to the most common A&P of humans that produce eggs.
- But wide variation exists and humans do not fall into binary sexes.



Application Question 12

Pretend you are synthesizing reproductive hormones as possible new ingredients in *female* contraceptive pills. You have so far successfully synthesized these two hormones:

--FSH

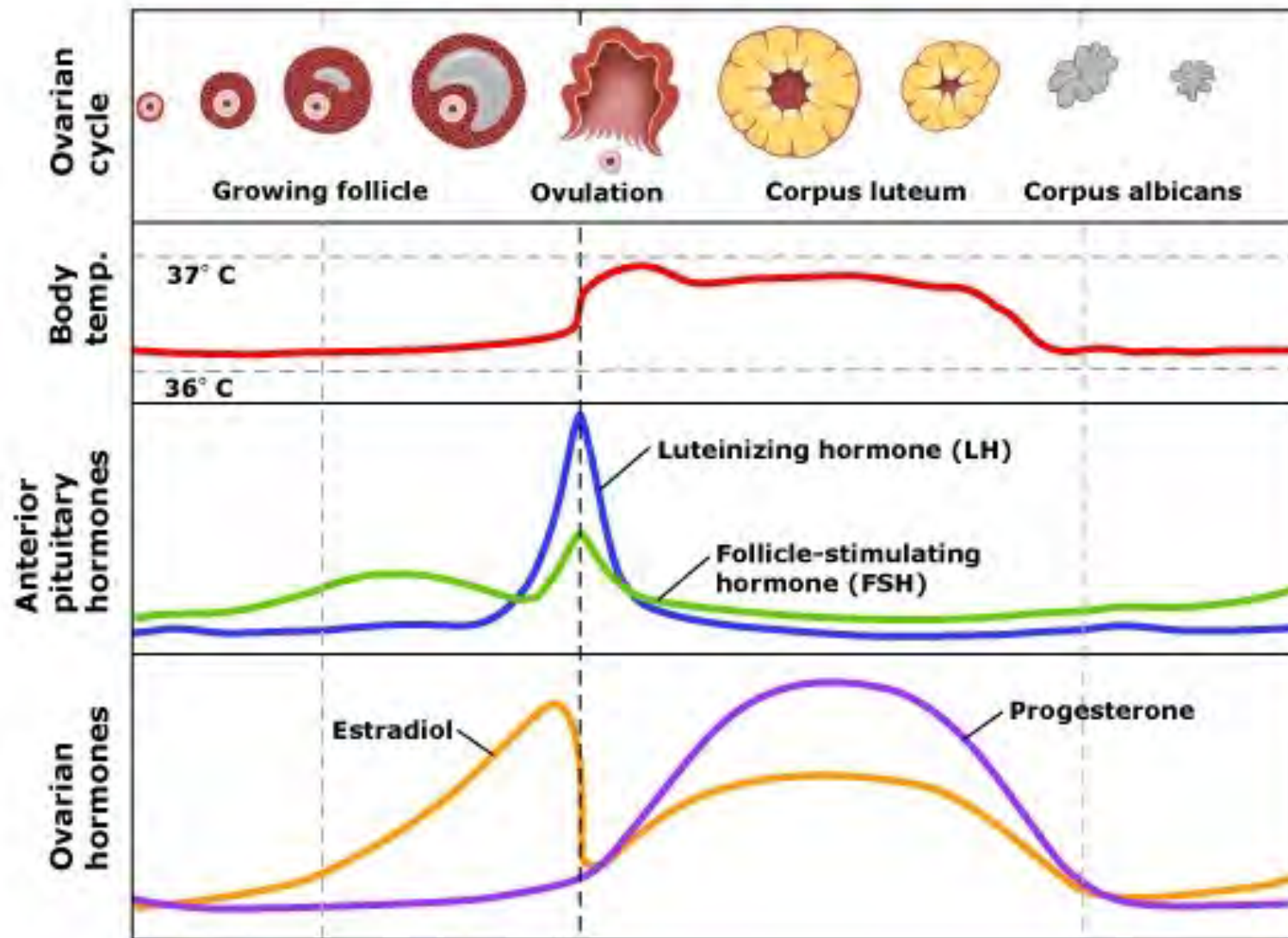
--Estrogen

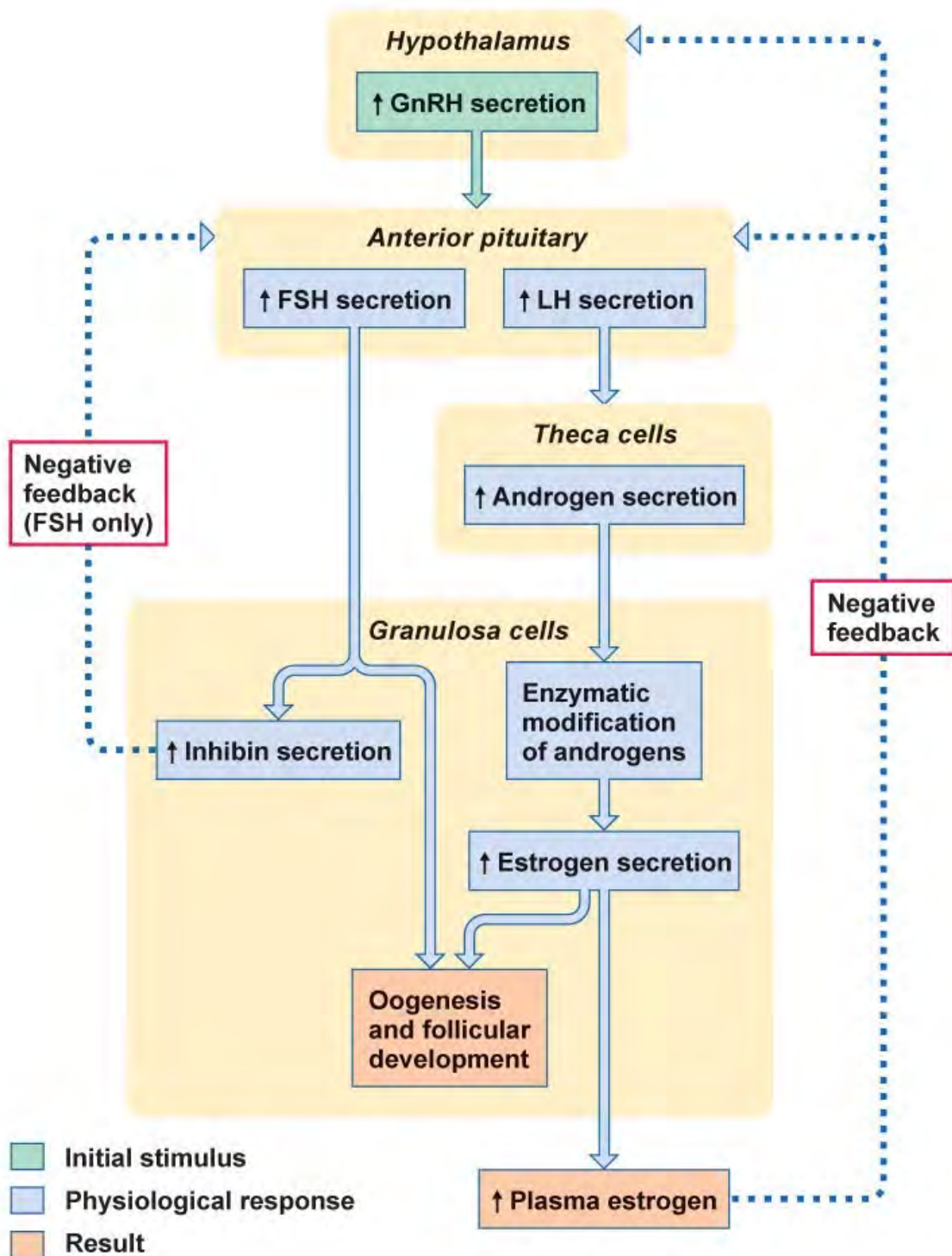
Given what you know about the roles of these hormones in females, do you think FSH and/or estrogen would be effective in preventing pregnancy? Why?

"FSH helps the egg develop and mature. Estrogen helps prepare the uterus for a fertilized egg to arrive. This means neither FSH or estrogen would be a good birth control therapy."

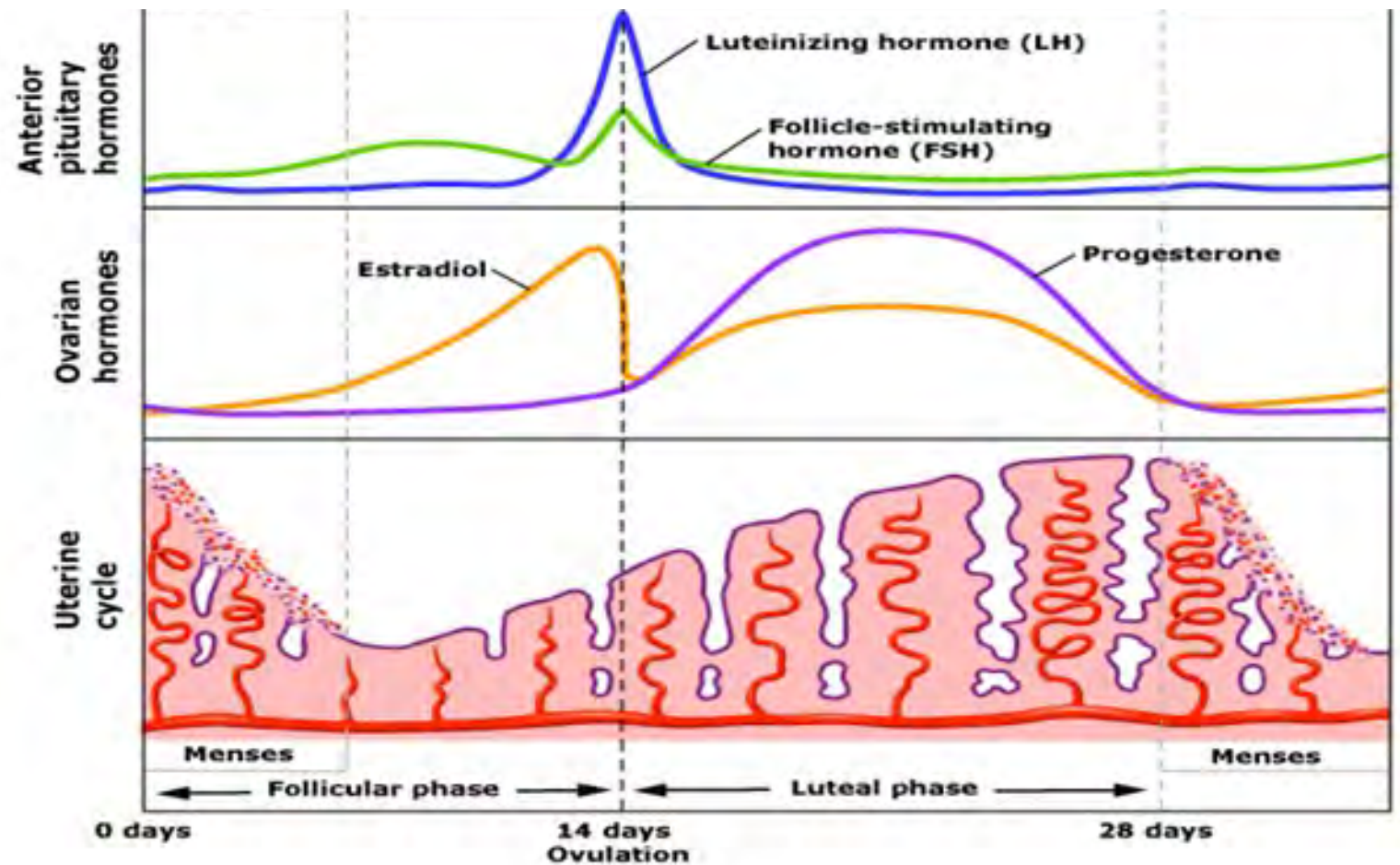
How could we make this more accurate and detailed?

Considering the roles of FSH and estrogen: The *ovarian* cycle

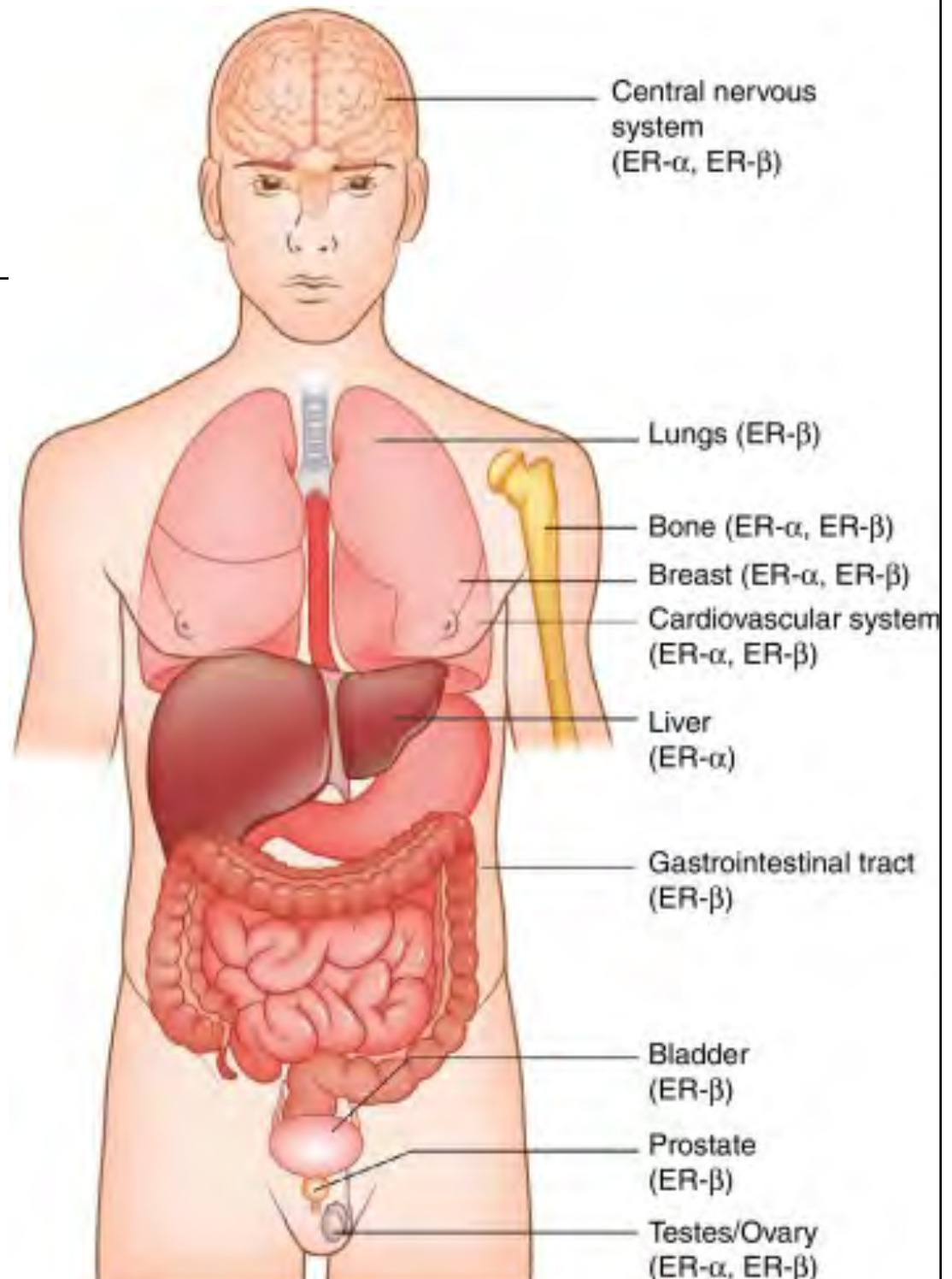




Considering the roles of FSH and estrogen: The *uterine* cycle



Estrogen Receptor Locations



What would this suggest about the potential side effects of birth control pills that include estrogen?

Why do birth control pills seem to be limited to females?



OPINION Are We Finally Ready for a Male Contraceptive?

Nov. 8, 2023



**Stay tuned! We'll look at
one example of male
chemical birth control
later!**

**First, we need to explore
a bit more about male
reproductive A&P...**



Recent News Regarding Local Basketball Star Draymond Green

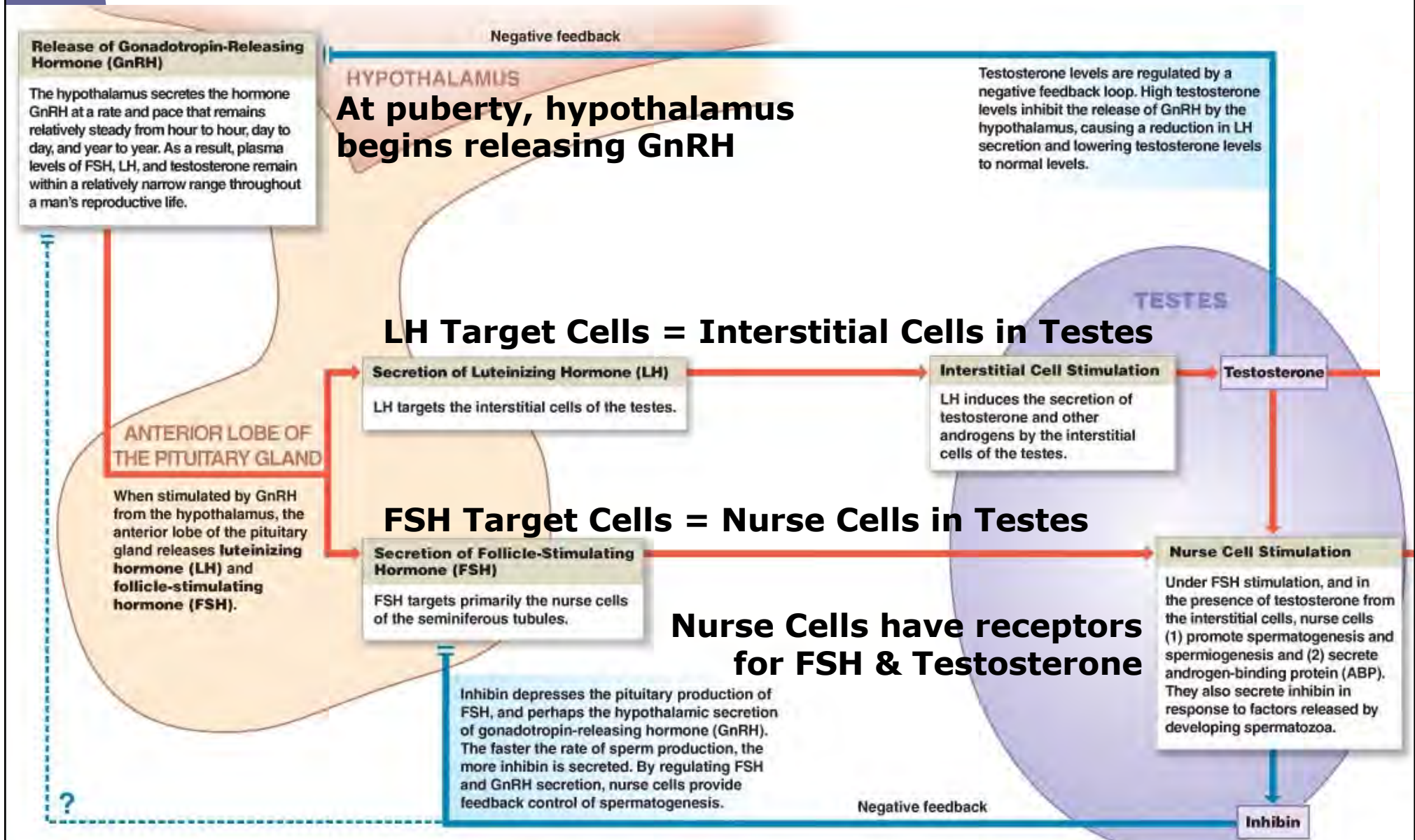


Draymond Green up to similar trouble back in 2016



Thinking as always through our “A&P perspective,” what reproductive system issues do you think could arise from an injury like the one happening in this picture?

Making Sperm: Stimulation of Sperm Creation



Making Sperm: FSH (and testosterone)

Stimulate Sperm Cell Divisions in Nurse Cells

Mitosis of spermatogonium

Each division of a diploid spermatogonium produces two daughter cells. One is a spermatogonium that remains in contact with the basal lamina of the tubule, and the other is a **primary spermatocyte** that is displaced toward the lumen.

Meiosis I

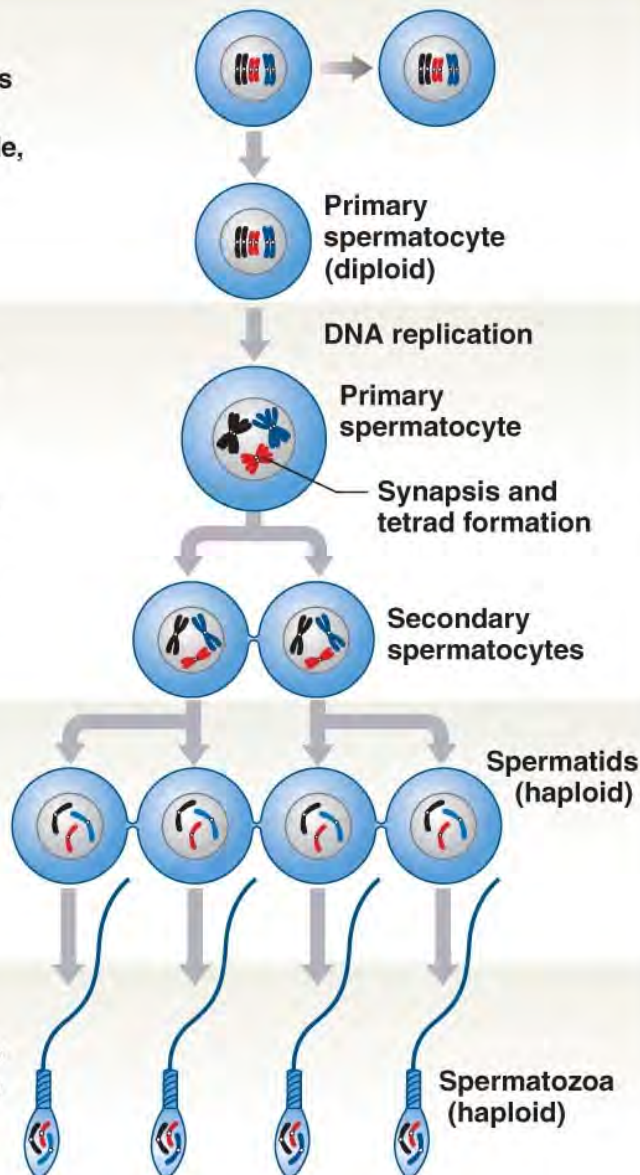
As meiosis I begins, each primary spermatocyte contains 46 individual chromosomes. At the end of meiosis I, the daughter cells are called **secondary spermatocytes**. Every secondary spermatocyte contains 23 chromosomes, each of which consists of a pair of duplicate chromatids.

Meiosis II

The secondary spermatocytes soon enter meiosis II, which yields four haploid **spermatids**, each containing 23 chromosomes. For each primary spermatocyte that enters meiosis, four spermatids are produced.

Spermiogenesis (physical maturation)

In spermiogenesis, the last step of spermatogenesis, each spermatid matures into a single spermatozoon, or sperm. The process of spermiogenesis takes roughly 5 weeks to complete.



Spermatogonia

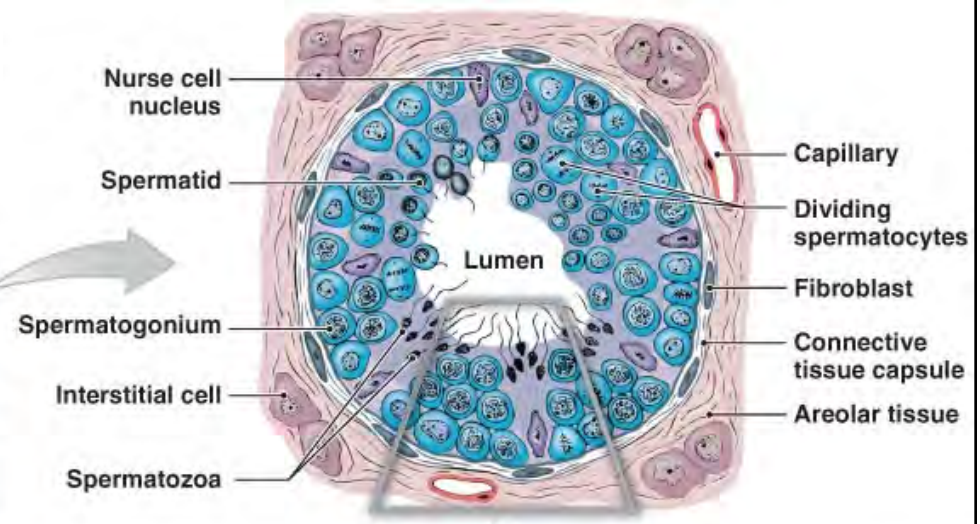
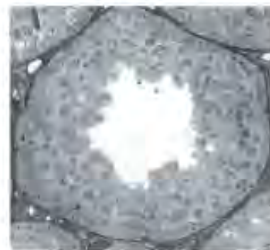
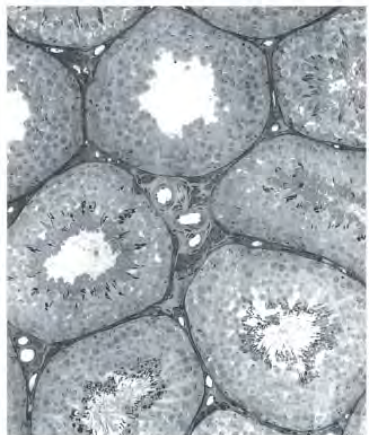
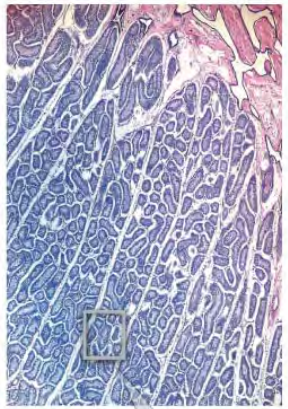
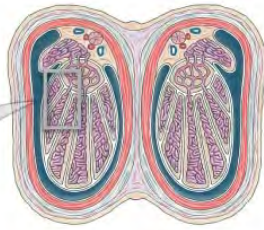
- Sperm stem cells
- Replenish during divisions

Spermatocytes

Spermatids

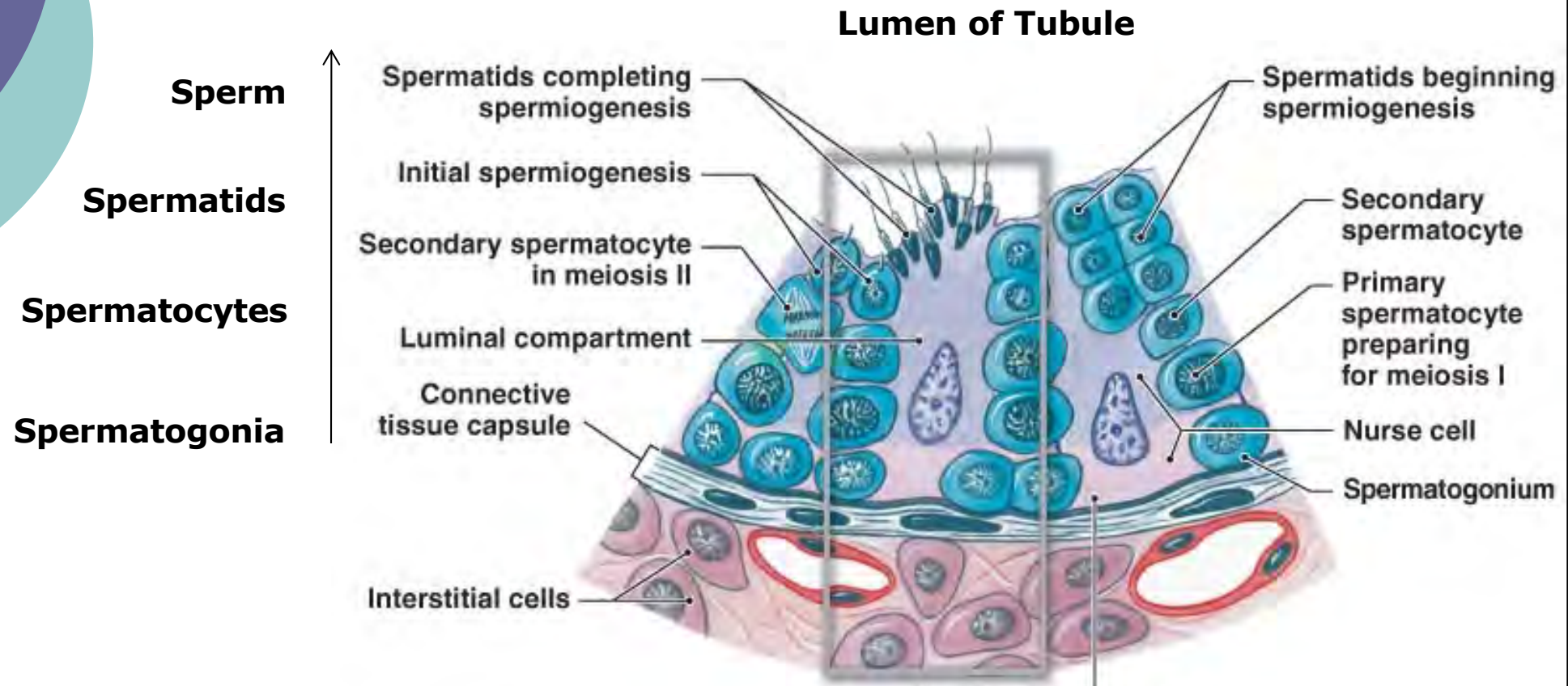
Sperm

Making Sperm: Nurse Cells are Located in Seminiferous Tubules

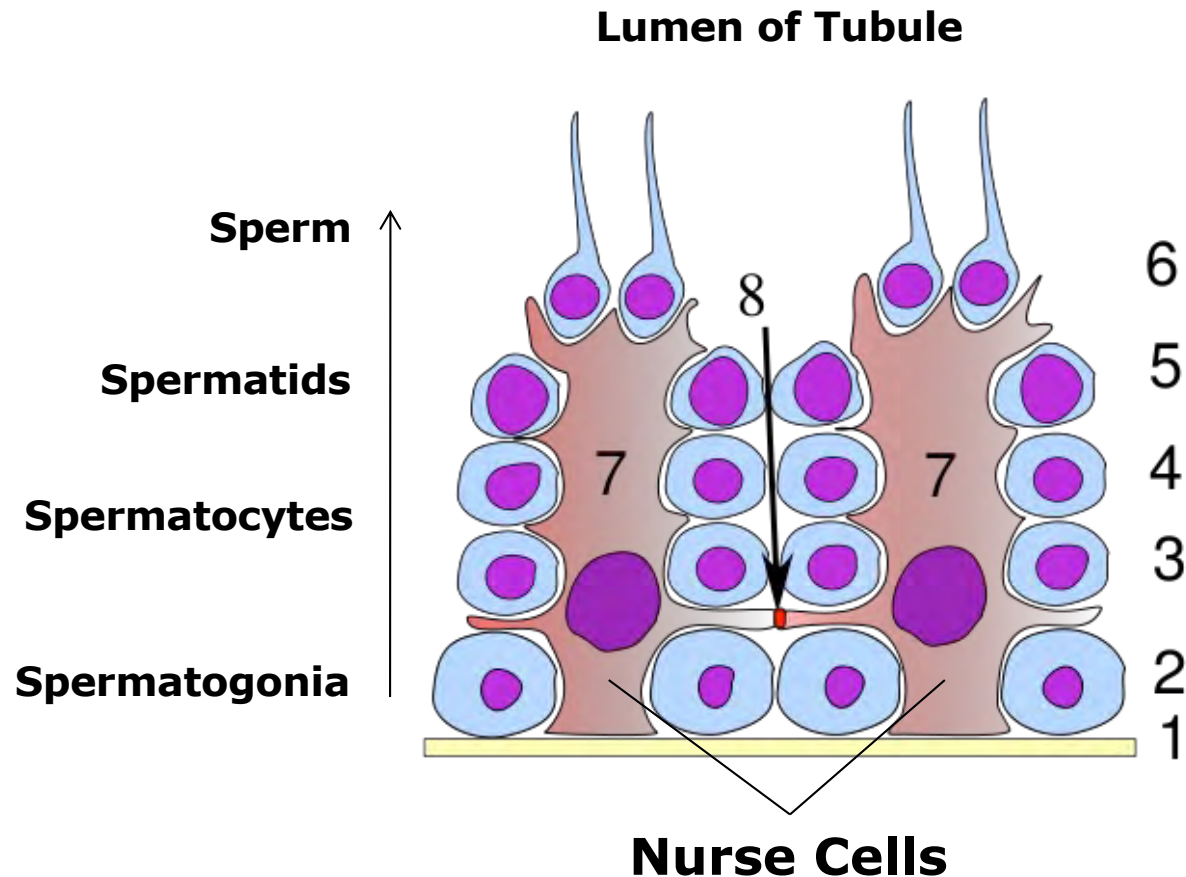


Seminiferous Tubules

Making Sperm: Stages of Development Begin on Outside of Seminiferous Tubules and End on Interior of Tubules

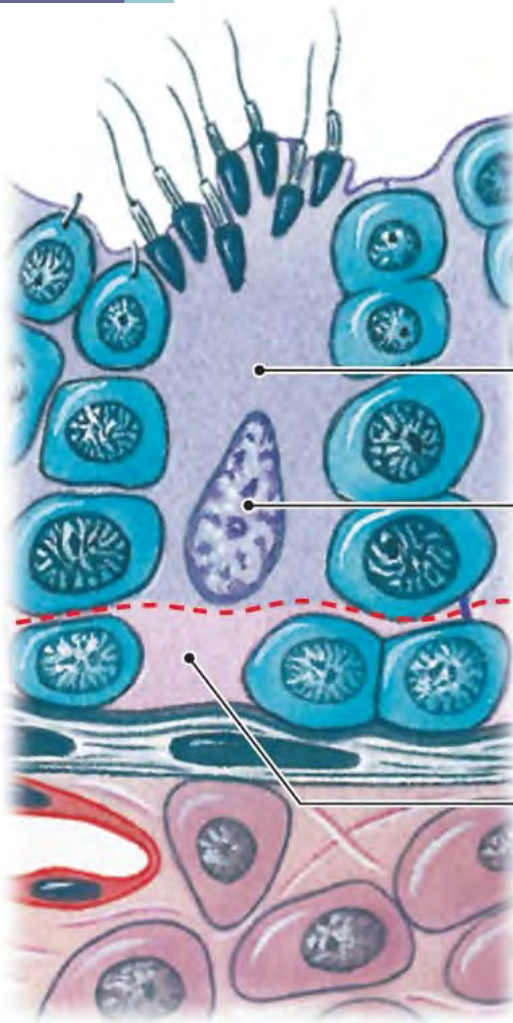


Making Sperm: Stages of Development Begin on Outside of Seminiferous Tubules and End on Interior of Tubules



Making Sperm

Lumen of Tubule

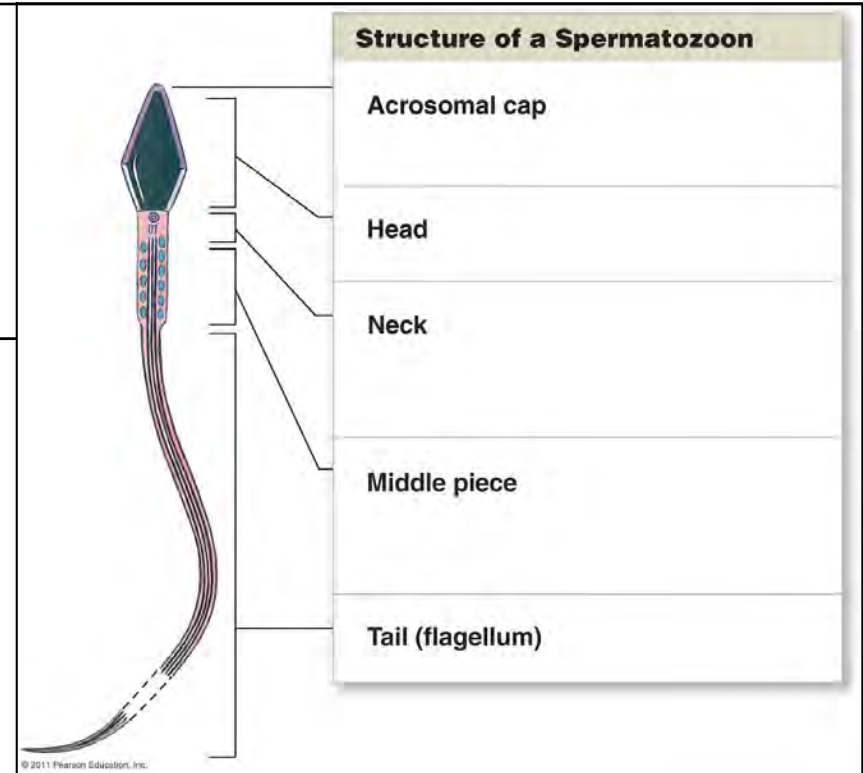


Luminal compartment

Nurse cell nucleus

Level of blood-testis barrier

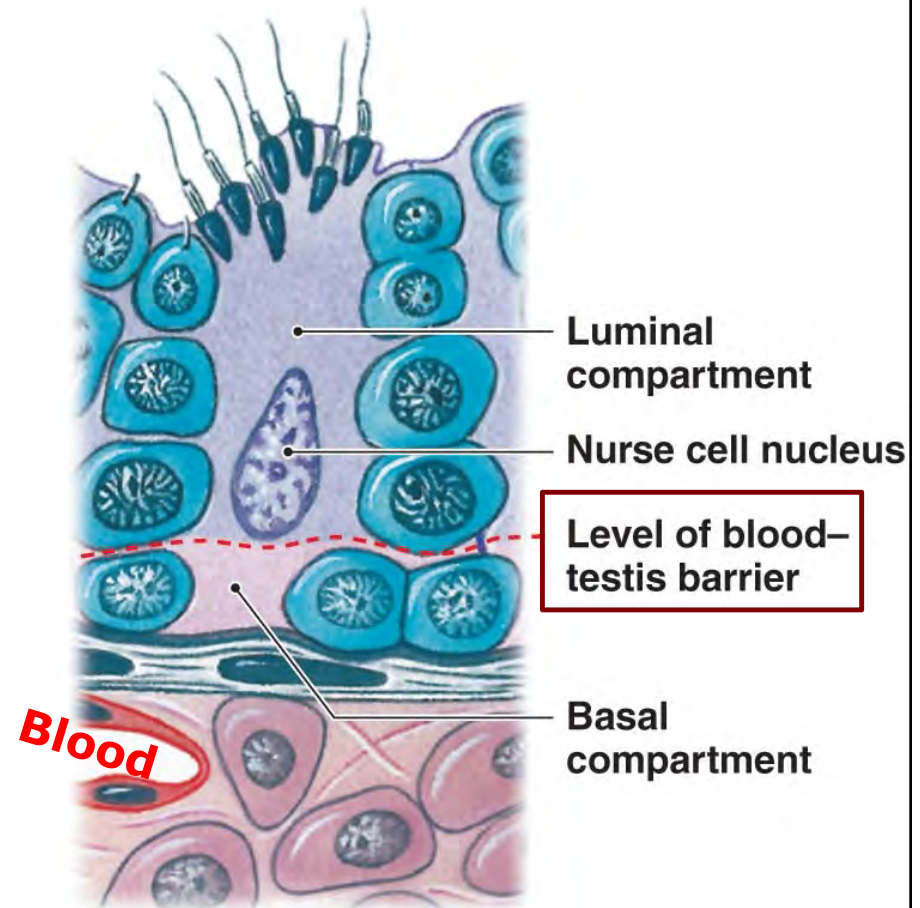
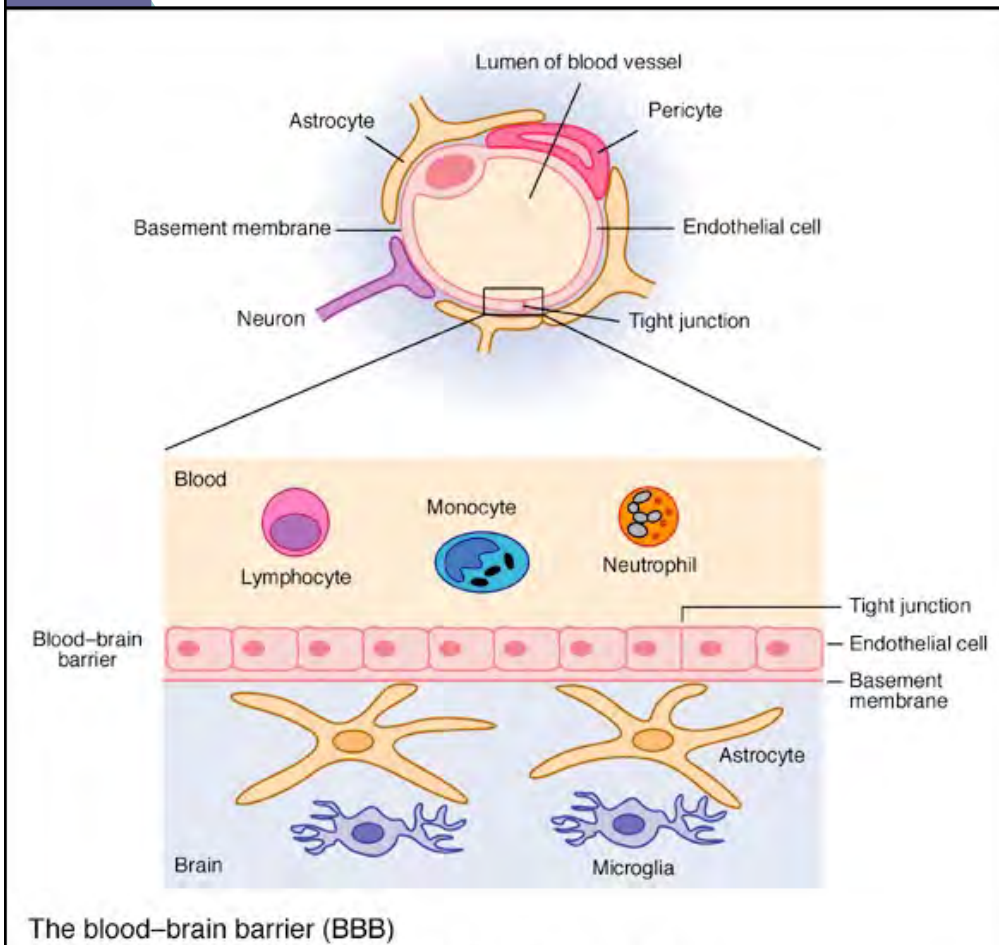
Basal compartment



Sperm end with acrosome pointing toward nurse cell/edge and tail pointing toward interior of tubule

Remember the blood brain barrier?

We also have a blood-testis barrier!

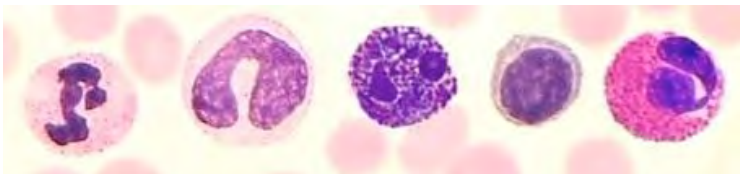


Nurse Cells Separate Sperm Cells from Blood Vessels

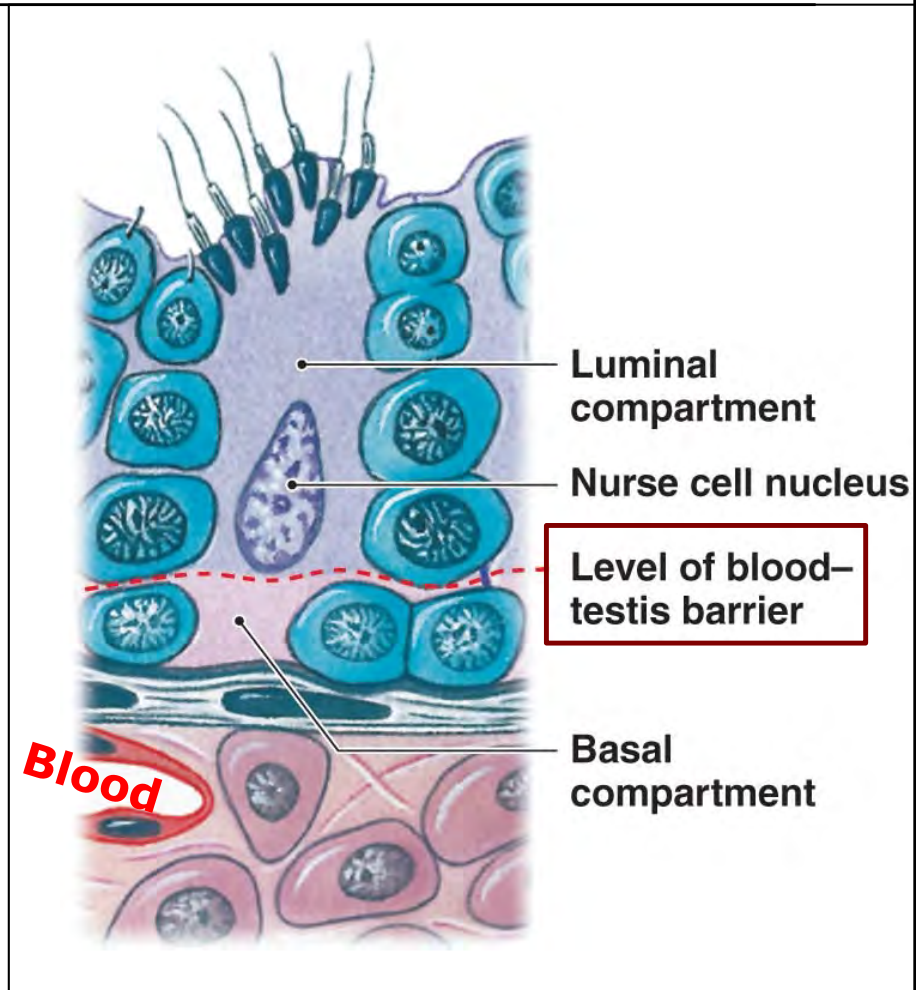
Why a blood-testis barrier?

Prevent harmful chemicals from interfering with spermatogenesis

Prevent white blood cells in blood from entering tubule



Sperm have membrane proteins that are recognized as foreign by WBCs



What do you think would happen if helper T cells and B cells were activated by sperm?



Thinking as always through our “A&P perspective,” what reproductive system issues do you think could arise from an injury like the one happening in this picture?

See if you can describe the involvement of Nurse Cells and white blood cells in this scenario.

13 **Your Name**

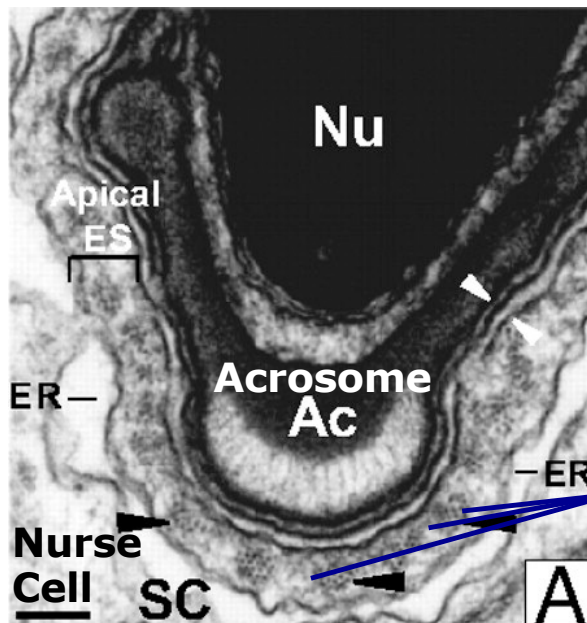
Your responses...

[illegible]

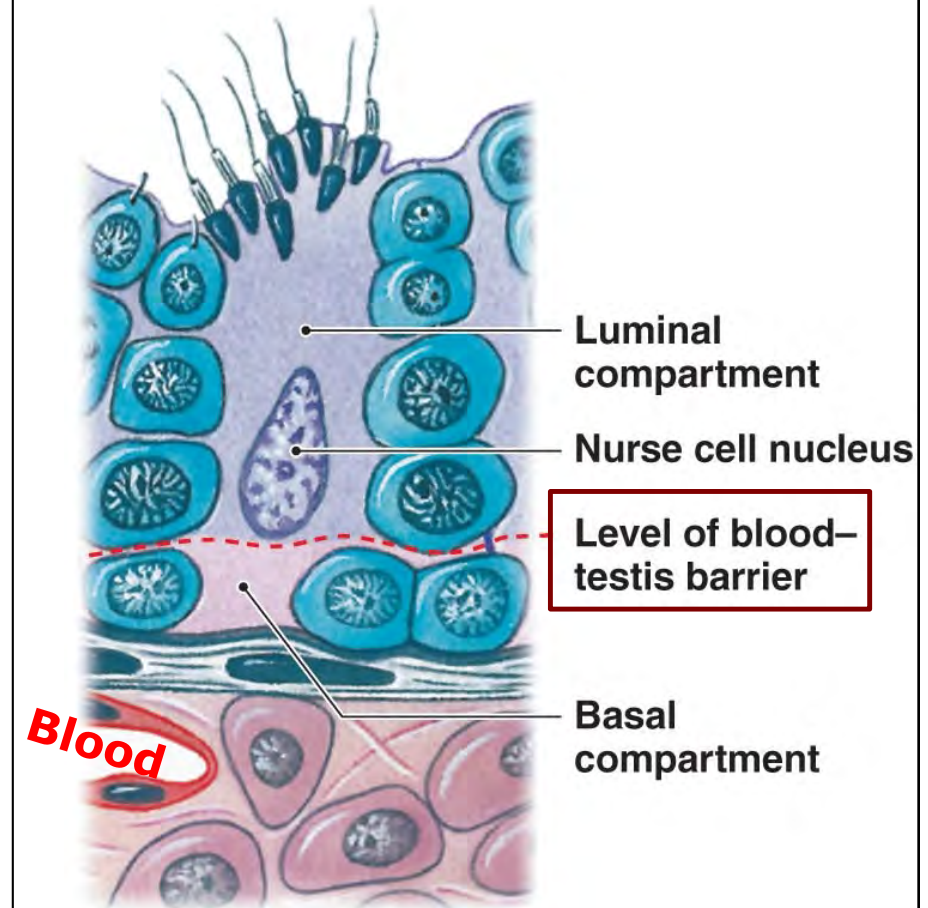
Problem with having a blood-testis barrier

Developing sperm still need nutrients from blood!

Solutions: Nurse cells supply nutrients to developing sperm. Developing cells are *physically attached* to nurse cells by surface proteins.

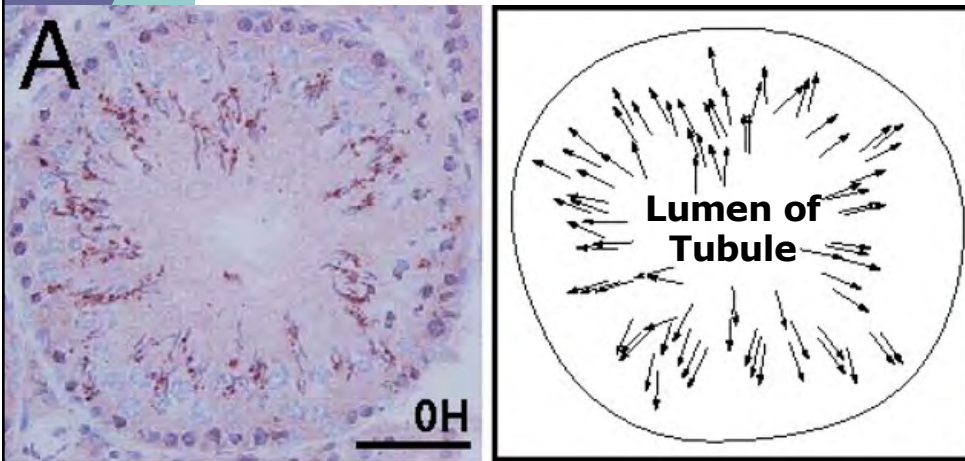


Actin Attachments



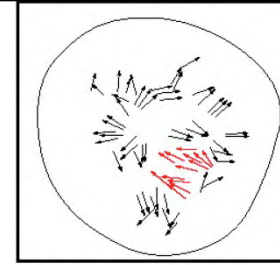
Adjudin: Male contraceptive that messes up attachment of developing cells to nurse cells

Picture and Drawing of a Seminiferous Tubule



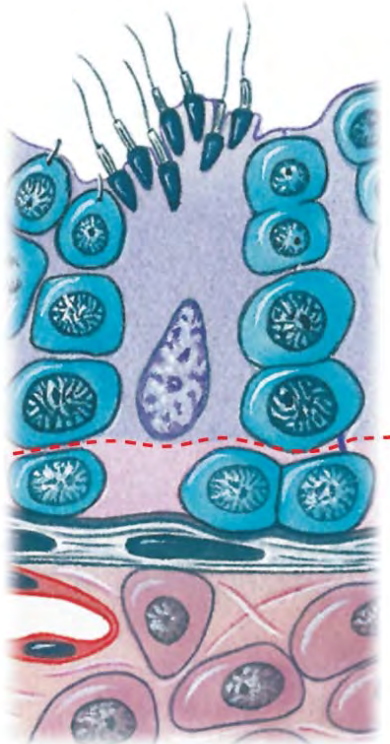
Before Adjudin: Sperm attached and tails extending toward lumen

Drawing it Out!

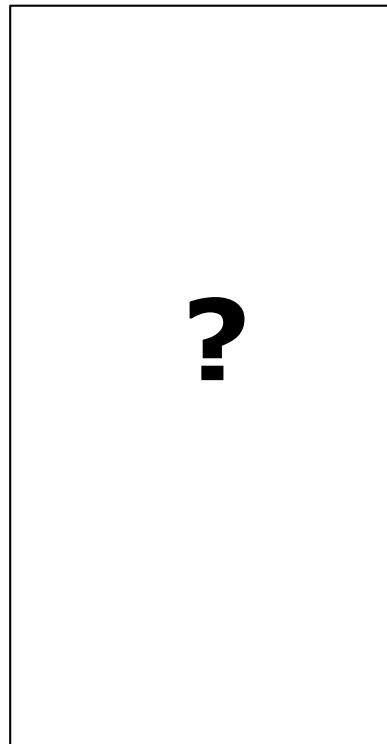


Work in groups of 2 (or no more than 3 if necessary). Choose to draw *either* **immune infertility** or contraception through **adjudin**. Use the general diagram format below, but change your diagram to show what happens in your version of infertility (immune or adjudin).

Normal



Your Version



Label cell names, the blood-testis barrier (in tact or not), and any other important features to show why infertility results under your conditions.